

Student Mental Health Data Analysis

Python Exploratory Data Analysis, Visualization and Logistic Regression Report

1. Introduction

This project presents a Python-based analysis of the Student Mental Health dataset. The work covers data cleaning, calculated columns, exploratory data analysis, sorting and filtering, index-based filtering, correlation analysis, descriptive statistics, Plotly visualizations, dashboard-style visuals, and Logistic Regression for predicting depression status.

2. Dataset Overview

The dataset contains 101 student records and 11 original variables, including gender, age, course, year of study, CGPA range, marital status, depression, anxiety, panic attack, and specialist treatment.

3. Data Cleaning

The dataset was checked for missing values and duplicate records. The CGPA field was standardized by removing an inconsistent trailing space. The data was then prepared for analysis and modelling.

4. Calculated Columns

Calculated column	Purpose
Age_Group	Groups students into 18–20, 21–23 and 24+.
Depression_Score	Encodes Depression Yes/No as 1/0.
Anxiety_Score	Encodes Anxiety Yes/No as 1/0.
Panic_Score	Encodes Panic Attack Yes/No as 1/0.
Mental_Health_Score	Sum of depression, anxiety and panic scores.
CGPA_Mid	Converts CGPA ranges to representative midpoint values.

5. Exploratory Data Analysis and Visualizations

Plotly was used to create six dashboard-style visuals: Gender Distribution; Number of Students by Course; Age Distribution; Students with Depression; Distribution of Mental Health Scores; and Average CGPA by Year of Study.

6. Sorting and Filtering

Five operations were performed: sorting Age descending, sorting CGPA midpoint descending, filtering Depression = Yes, filtering Anxiety = Yes, and filtering specialist treatment = Yes.

7. Index-Based Filtering

Seven operations were performed using Pandas `.iloc[]` and `.loc[]`, including row ranges, column selection, last-row selection, age filtering, female-student filtering, and Mental Health Score filtering.

8. Correlation Analysis

A labelled Plotly correlation heatmap was created using Age, CGPA_Mid, Depression_Score, Anxiety_Score, Panic_Score, and Mental_Health_Score. Age showed very weak relationships with the other variables. Depression, Anxiety and Panic showed positive relationships with the combined Mental_Health_Score. These strong component-to-total relationships are expected because the total score is constructed from those components.

Selected correlations were approximately: Age–CGPA 0.03; Age–Depression –0.067; Age–Anxiety –0.089; Age–Panic 0.06; Depression–Anxiety 0.274; Depression–Panic 0.247; Anxiety–Panic 0.084; Depression–Mental Health Score 0.743; Anxiety–Mental Health Score 0.652; Panic–Mental Health Score 0.646.

9. Descriptive Statistics

Descriptive statistics were included to summarize numerical variables using measures such as count, mean, standard deviation, minimum, quartiles and maximum, with categorical summaries also examined.

10. Logistic Regression

Logistic Regression was used to predict whether a student reported depression. Depression was encoded as Yes = 1 and No = 0. The predictors were Age, CGPA_Mid, Anxiety_Score and Panic_Score. An 80/20 stratified train-test split was used.

Model result	Value
Test observations	21
Accuracy	71.43%
Correct predictions	15 of 21
Confusion matrix	[[14, 0], [6, 1]]
Recall – No Depression	100%
Recall – Depression	14%

11. Model Interpretation

The Logistic Regression model achieved 71.43% accuracy. The confusion matrix showed 14 correctly identified No Depression cases and 1 correctly identified Depression case. Six students who actually reported depression were classified as No Depression. The classification report therefore shows much stronger performance for class 0 than class 1. The relatively small dataset and test set should be considered when interpreting these results.

12. Linear Regression and Logistic Regression

Linear Regression predicts a continuous numerical outcome. A possible use case is predicting a student's CGPA from relevant numerical characteristics.

Logistic Regression predicts the probability of a categorical outcome, commonly a binary Yes/No outcome. In this project it was used to predict whether a student reported depression.

13. Conclusion

This project demonstrates a complete Python data-analysis workflow using a real-world student mental-health dataset. It combines data preparation, feature engineering, exploratory analysis, visualization, statistical description and predictive modelling. The Logistic Regression model achieved 71.43% accuracy on the test set, while its 14% recall for Depression cases indicates that the model had difficulty identifying positive cases. Overall, the project demonstrates practical use of Python, Pandas, Plotly and Scikit-learn for data analysis.

14. Tools and Technologies

Python, Pandas, Plotly and Scikit-learn were used for the analysis in a Jupyter/Colab-style notebook environment.